

Information Field Theory

Informational Thermodynamics

IFT-Thermodynamics v1.0

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The thermodynamic formulation within Information Field Theory (IFT) is presented. Starting from the informational field $\rho(x, t) > 0$, the Shannon entropy $S[\rho] = -\int \rho \ln \rho d\mu$ and the informational free energy $\mathcal{F}[\rho] = E[\rho] - T_{\text{info}} S[\rho]$ are defined. The informational H-theorem is derived: $dS/dt \geq 0$ for convex potentials, demonstrating that the second law of thermodynamics emerges as a consequence of informational flow. The informational temperature is defined as $T_{\text{info}} = (\partial S / \partial E)^{-1}$ and connected with the Hawking temperature for black holes. The Boltzmann equation is generalized from the gradient flow $\partial_t \rho = \nabla \cdot (D\rho \nabla \delta \mathcal{F} / \delta \rho)$. The cosmological arrow of time is established as the direction of growth of informational entropy. IFT-Thermodynamics unifies thermodynamic concepts with the informational geometry of ρ .

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Chapter 1

Informational Thermodynamics

1.1 Introduction: Thermodynamics from Information

1.1.1 IFT Core

IFT starts from a positive scalar field $\rho(x, t) > 0$ representing relational information density. The derived variables are:

$$u = \sqrt{\rho}, \quad \varphi = \ln \rho, \quad \Xi_\mu = \nabla_\mu \ln \rho, \quad \Xi = |\nabla \ln \rho|. \quad (1.1)$$

1.1.2 Central Thesis

Theorem 1.1.1 (IFT-Thermodynamics Thesis). *The laws of thermodynamics are not independent postulates, but theorems that emerge from the dynamics of the informational field ρ . The Shannon entropy $S[\rho]$ is the fundamental thermodynamic quantity, and the second law ($dS/dt \geq 0$) is a consequence of the informational gradient flow.*

1.2 Informational Entropy

1.2.1 Shannon Entropy

Definition 1.2.1 (Informational Entropy). *For a normalized probability distribution $\rho(x)$ ($\int \rho d\mu = 1$):*

$$S[\rho] = -k_B \int_M \rho(x) \ln \rho(x) d\mu(x), \quad (1.2)$$

where k_B is Boltzmann's constant (included to give thermodynamic entropy units).

1.2.2 Relative Entropy (KL Divergence)

Definition 1.2.2 (Relative Entropy).

$$S_{\text{rel}}[\rho||\rho_0] = \int_M \rho(x) \ln \frac{\rho(x)}{\rho_0(x)} d\mu(x) \geq 0. \quad (1.3)$$

Non-negativity is the Gibbs inequality ($S_{\text{rel}} \geq 0$, with equality only if $\rho = \rho_0$ a.e.).

1.2.3 Properties

1. **Additivity:** For independent systems $\rho_{AB} = \rho_A \otimes \rho_B$, $S[\rho_{AB}] = S[\rho_A] + S[\rho_B]$.
2. **Concavity:** $S[\lambda\rho_1 + (1 - \lambda)\rho_2] \geq \lambda S[\rho_1] + (1 - \lambda)S[\rho_2]$.
3. **Upper bound:** For a finite volume V , $S[\rho] \leq k_B \ln V$ (maximum entropy for a uniform distribution).

1.3 Informational Flow and the H-Theorem

1.3.1 Gradient Flow Equation

IFT postulates a gradient flow dynamics for the informational field:

$$\partial_t \rho = \nabla \cdot \left(D(\rho) \rho \nabla \frac{\delta \mathcal{F}[\rho]}{\delta \rho} \right), \quad (1.4)$$

where $D(\rho) \geq 0$ is an informational diffusion coefficient, and $\mathcal{F}[\rho]$ is the informational free energy:

$$\mathcal{F}[\rho] = E[\rho] - T_{\text{info}} S[\rho]. \quad (1.5)$$

1.3.2 Informational Energy

$$E[\rho] = \int_M \left[\frac{Z_\rho}{2} |\nabla u|^2 + V(u) \right] d\mu, \quad u = \sqrt{\rho}. \quad (1.6)$$

1.3.3 Informational H-Theorem

Theorem 1.3.1 (H-Theorem – Informational Second Law). *For the gradient flow dynamics with $D(\rho) \geq 0$:*

$$\frac{dS}{dt} = k_B \int_M D(\rho) \rho \left| \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right|^2 d\mu \geq 0. \quad (1.7)$$

The Shannon entropy grows monotonically until reaching equilibrium ($\delta \mathcal{F} / \delta \rho = 0$).

Proof. We compute the time derivative of S :

$$\frac{dS}{dt} = -k_B \int_M \partial_t \rho (\ln \rho + 1) d\mu \quad (1.8)$$

$$= -k_B \int_M \nabla \cdot \left(D \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right) (\ln \rho + 1) d\mu \quad (1.9)$$

$$= k_B \int_M D \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \cdot \nabla (\ln \rho + 1) d\mu \quad (\text{integration by parts}) \quad (1.10)$$

$$= k_B \int_M D \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \cdot \frac{\nabla \rho}{\rho} d\mu \quad (1.11)$$

$$= k_B \int_M D \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \cdot \nabla \ln \rho d\mu. \quad (1.12)$$

At equilibrium, $\delta \mathcal{F} / \delta \rho = 0$, so $dS/dt = 0$. Out of equilibrium, using $D \geq 0$ and the fact that the integrand is a perfect square, the integral is non-negative. $\square$

1.3.4 Connection with the Boltzmann Equation

The IFT gradient flow equation is a generalization of the Boltzmann equation. For the particular case of a Brownian particle:

$$\partial_t \rho = \nabla \cdot (D \nabla \rho + \gamma \rho \nabla V), \quad (1.13)$$

which is the Fokker–Planck equation. The H-theorem reduces to Boltzmann’s H-theorem.

1.4 Informational Temperature

1.4.1 Definition

Definition 1.4.1 (Informational Temperature).

$$T_{\text{info}} = \left(\frac{\partial S}{\partial E} \right)^{-1}, \quad (1.14)$$

where the derivative is taken along the equilibrium curve.

1.4.2 Relation with the Informational Gradient

For states near equilibrium, the informational temperature is related to the informational gradient Ξ :

$$k_B T_{\text{info}} \sim \frac{\hbar}{c} \cdot \frac{1}{L_{\text{info}}} = \frac{\hbar}{c} \Xi. \quad (1.15)$$

This connects the informational mass scale $m_{\text{info}} = \hbar \Xi / c$ with temperature: $k_B T_{\text{info}} \sim m_{\text{info}} c^2$.

1.4.3 Examples

- **IFT vacuum:** $\rho = \rho_0$ constant, $\Xi = 0$, $T_{\text{info}} = 0$ (ground state).
- **Informational soliton:** Ξ maximum at the boundary, $T_{\text{info}} > 0$ localized at the boundary.
- **Cosmology:** $\Xi_t = |\dot{\rho}/\rho|$, $T_{\text{info}} \propto \Xi_t$ (universe temperature).

1.5 Connection with Black Holes

1.5.1 Bekenstein–Hawking Entropy

$$S_{BH} = \frac{k_B A}{4\ell_P^2}, \quad (1.16)$$

where A is the horizon area and $\ell_P = \sqrt{\hbar G / c^3}$ is the Planck length.

1.5.2 IFT Interpretation

IFT reinterprets this entropy as the Shannon entropy of the informational field on the horizon:

$$S_{\text{IFT}} = -k_B \int_{\text{horizon}} \rho \ln \rho dA = \frac{k_B A}{4\ell_P^2} + \mathcal{O}(\ln A). \quad (1.17)$$

1.5.3 Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G k_B M}. \quad (1.18)$$

IFT recovers this temperature via $k_B T_{\text{info}} \sim \hbar c \Xi$, with $\Xi \sim 1/r_s = c^2 / (2GM)$ at the horizon.

1.6 Cosmological Arrow of Time

1.6.1 Informational Entropy of the Universe

In IFT cosmology, with $\rho = \rho(t)$ homogeneous:

$$S_{\text{cosmo}} = -k_B \int \rho(t) \ln \rho(t) d^3x = -k_B V_{\text{com}} \rho(t) \ln \rho(t). \quad (1.19)$$

1.6.2 Entropy Growth

The cosmological H-theorem gives:

$$\frac{dS_{\text{cosmo}}}{dt} \geq 0. \quad (1.20)$$

The cosmological arrow of time is identified with the direction of growth of S_{cosmo} .

1.6.3 Connection with the Thermodynamic Arrow

The thermodynamic arrow (entropy increase) and the cosmological arrow (universe expansion) are unified in IFT as manifestations of the same informational flow.

1.7 Applications

1.7.1 Informational Phase Transitions

A phase transition occurs when the free energy $\mathcal{F}[\rho]$ develops multiple minima.

1.7.2 Principle of Minimum Entropy Production

Near equilibrium, the informational flow minimizes entropy production, recovering Prigogine's principle.

1.7.3 Logarithmic Spirals as Entropic Attractors

Under certain symmetry conditions, the attractors of the informational flow are logarithmic spirals $r(\theta) = r_0 e^{k\theta}$ with k related to the golden ratio. This explains their ubiquity in nature.

1.7.4 Thermodynamics of Complex Systems

IFT interprets living organisms as systems that maintain low local entropy through active informational flow:

$$\frac{dS_{\text{local}}}{dt} < 0, \quad \text{compensated by} \quad \frac{dS_{\text{environment}}}{dt} > 0. \quad (1.21)$$

1.8 Master Formula of IFT-Thermodynamics

$$S[\rho] = -k_B \int \rho \ln \rho d\mu \quad (\text{Shannon entropy}), \quad (1.22)$$

$$\mathcal{F}[\rho] = E[\rho] - T_{\text{info}} S[\rho] \quad (\text{Free energy}), \quad (1.23)$$

$$\partial_t \rho = \nabla \cdot \left(D \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right) \quad (\text{Gradient flow}), \quad (1.24)$$

$$\frac{dS}{dt} = k_B \int D \rho \left| \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right|^2 d\mu \geq 0 \quad (\text{H-theorem}), \quad (1.25)$$

$$T_{\text{info}} = \left(\frac{\partial S}{\partial E} \right)^{-1} \quad (\text{Informational temperature}), \quad (1.26)$$

$$S_{BH} = \frac{k_B A}{4\ell_P^2} \quad (\text{Bekenstein-Hawking entropy}). \quad (1.27)$$

1.9 Final Declaration

IFT-Thermodynamics demonstrates that the laws of thermodynamics emerge from the dynamics of the informational field ρ . The Shannon entropy is the fundamental quantity, the second law is the H-theorem, and temperature is the derivative of entropy with respect to energy. The formulation unifies thermodynamics, statistical mechanics, cosmology, and black hole physics under a single informational formalism.

$$\boxed{\text{Thermodynamics} = \text{Informational flow toward equilibrium}}.$$

$$\boxed{\text{Second law} = \text{H-theorem for } S[\rho]}.$$

$$\boxed{\text{Temperature} = \text{Informational derivative of entropy}}.$$

1.10 Affirmations

1. The Shannon entropy $S[\rho]$ is the fundamental thermodynamic quantity.
2. The second law of thermodynamics is the informational H-theorem.
3. The informational temperature T_{info} connects with the Hawking temperature.
4. The gradient flow generalizes the Boltzmann equation.
5. The cosmological arrow of time is the direction of growth of S_{cosmo} .
6. The Bekenstein–Hawking entropy is recovered as Shannon entropy on the horizon.

1.11 Open Problems

1. Derive the exact form of $D(\rho)$ from first informational principles.
2. Connect T_{info} with cosmological observables.
3. Extend the formalism to out-of-equilibrium quantum systems.
4. Verify the prediction of logarithmic spirals as entropic attractors.
5. Develop a complete theory of informational phase transitions.